

Index

SL NO	ASSIGNMENT NAME	DATE OF EXPERIMENT	PAGE NO
01	Write a Lex Program to Check Whether a Given Number is Odd or Even		01
02	Write a Lex Program to Find the Area of a Circle		02
03	Write a Lex Program to Find the Sum of N Numbers		03
04	Write a Lex Program to Find the Sum of First N Natural Numbers		04
05	Write a Lex Program to Count the Total Number of Vowels in a String		05
06	Write a Lex Program to Count the Number of Words in a Line		06
07	Write a Lex Program to Recognize English Words According to English Grammar Categories		07
08	Write a Lex Program to Count Total Words, Lines, Characters, Numbers and Special Characters		08
09	Write a Lex Program to Find the Factorial of a Given Number		09

Teacher's signature

❖ **Assignment No: 1**

❖ **Assignment Name:** Write a Lex program to check whether a given number is Odd or Even?

▪ **Source Code:**

```
%{
    #include <stdio.h>
    #include <stdlib.h>
    int i;
}%

%%
[0-9]+
{
    i = atoi(yytext);

    if(i % 2 == 0)
        printf("Even Number\n");
    else
        printf("Odd Number\n");
}

%%

int main ()
{
    printf("Enter a number: ");
    yylex();
    return 0;
}

int yywrap()
{
    return 1;
}
```

▪ **Output:**

Enter a Number: 4
Even Number

❖ **Assignment No: 2**

❖ **Assignment Name:** Write a Lex Program to Find the Area of a Circle?

▪ **Source Code:**

```
%{
#include <stdio.h>
#include <stdlib.h>
float r;
#define pi 3.14159
}%
%%
([0-9]*[.])?[0-9]+
{
    r = atof(yytext);
    printf("Area of Circle = %.6f\n", pi * r * r);
}
.|\\n
{
    printf("Wrong! Not a Number\n");
}

%%

int main()
{
    printf("Enter the Radius: ");
    yylex();
    return 0;
}
int yywrap()
{
    return 1;
}
```

▪ **Output:**

```
Enter the Radius: 5
Area of Circle = 78.539749
```

❖ **Assignment No: 3**

❖ **Assignment Name:** Write a Lex Program to Find the Sum of N Numbers?

▪ **Source Code:**

```
%{  
    #include <stdio.h>  
    #include <stdlib.h>  
    int op = 0;  
    float a, b;  
    void digi();  
}%  
%%  
[0-9]+|([0-9]*)."([0-9]+)  { digi(); }  
  
"+" { op = 1; }  
  
"\n"{ printf("\n Result : %f\n", a); }  
%%  
void digi()  
{  
    if(op == 0)  
        a = atof(yytext);  
    else  
    {  
        b = atof(yytext);  
        a = a + b;  
    }  
}  
int main()  
{  
    printf("Enter Numbers:\n");  
    yylex();  
    return 0;  
}  
int yywrap()  
{  
    return 1;  
}
```

▪ **Output:**

```
Enter Numbers:  
10+20+30  
Result :60.000000
```

❖ **Assignment No: 4**

❖ **Assignment Name:** Write a Lex Program to Find the Sum of First N Natural Numbers?

❖ **Source Code:**

```
%{  
    #include <stdio.h>  
    #include <stdlib.h>  
  
    int i;  
}%  
  
%%  
[0-9]+  
{  
    i = atoi(yytext);  
    printf("The sum of first %d number is: %d\n", i, (i * i + i) / 2);  
}  
  
%%  
  
int main()  
{  
    printf("enter the last number:\n");  
    yylex();  
    return 0;  
}  
  
int yywrap()  
{  
    return 1;  
}
```

❖ **Ouput:**

```
enter the last number:  
10  
The sum of first 10 number is:55
```

❖ **Assignment No: 5**

❖ **Assignment Name:** Write a Lex Program to Count the Total Number of Vowels in a String?

❖ **Source Code:**

```
%{  
    #include <stdio.h>  
    int v = 0;  
}%  
  
%%  
[aeiouAEIOU] { v++; }  
[a-zA-Z] { }  
[\n]{ return 0; }  
  
%%  
  
int main()  
{  
    printf("Enter the string:\n");  
    yylex();  
  
    printf("Number of vowels are: %d\n", v);  
  
    return 0;  
}  
  
int yywrap()  
{  
    return 1;  
}
```

❖ **Output:**

```
Enter the string:  
HELLO  
Number of vowels are: 2
```

❖ **Assignment No: 6**

❖ **Assignment Name:** Write a Lex Program to Count the Number of Words in a Line?

▪ **Source Code:**

```
%{  
    #include <stdio.h>  
    #include <string.h>  
    int i = 0;  
}%  
  
%%  
[a-zA-Z0-9]+ { i++; }  
  
"\n"{ printf("The number of words is %d\n", i); }  
  
[ \t] { }  
  
%%  
int main()  
{  
    printf("Enter the line:\n");  
    yylex();  
    return 0;  
}  
  
int yywrap()  
{  
    return 1;  
}
```

▪ **Output:**

```
Enter the line:  
Hello World  
The number of words is 2
```

❖ **Assignment No: 7**

- ❖ **Assignment Name:** Write a Lex Program to Recognize English Words According to English Grammar Categories (Auxiliary Verb, Adverb, Preposition, Conjunction, Pronoun, and Noun)?

▪ **Source Code:**

```
%{  
    #include <stdio.h>  
}%  
%%  
(?i:am)|(?i:is)|(?i:are)  
    { printf("Auxiliary Verb\n"); }  
  
(?i:very)|(?i:simply)  
    { printf("Adverb\n"); }  
  
(?i:to)|(?i:from)|(?i:behind)  
    { printf("Preposition\n"); }  
  
(?i:if)|(?i:then)|(?i:and)|(?i:but)  
    { printf("Conjunction\n"); }  
  
(?i:i)|(?i:you)|(?i:he)|(?i:she)  
    { printf("Pronoun\n"); }  
  
[a-zA-Z]+  
    { printf("Noun\n"); }  
%%  
int main()  
{  
    printf("Enter the word:\n");  
    yylex();  
    return 0;  
}  
int yywrap()  
{  
    return 1;  
}
```

▪ **Output:**

```
Enter the word:  
Am  
Auxiliary Verb
```

❖ **Assignment No: 8**

❖ **Assignment Name:** Write a Lex Program to Count Total Words, Lines, Characters, Numbers, and Special Characters from the Input?

▪ **Source Code:**

```
%{
    #include <stdio.h>
    int line = 0, word = 0, letter = 0, number = 0, splchr = 0;
}%
%%
"\n" { line++; word++; }
[\t ] { word++; }

[a-zA-Z]{ letter++; }

[0-9] { number++; }

. { splchr++; }

%%
int main()
{
    printf("Enter the input:\n");
    yylex();
    printf("Total lines = %d\n", line);
    printf("Total words = %d\n", word);
    printf("Total letters = %d\n", letter);
    printf("Total numbers = %d\n", number);
    printf("Total special characters = %d\n", splchr);
    return 0;
}
int yywrap()
{
    return 1;
}
```

▪ **Output:**

```
Hello World 123
Total lines = 1
Total words = 3
Total letters = 10
Total numbers = 3
Total special characters = 0
```

❖ **Assignment No: 9**

❖ **Assignment Name:** Write a Lex Program to Find the Factorial of a Given Number?

▪ **Source Code:**

```
%{  
    #include <stdio.h>  
    #include <stdlib.h>  
    int n, i, f = 1;  
%}  
%%  
[0-9]+  
{  
    n = atoi(yytext);  
  
    for(i = 1; i <= n; i++)  
    {  
        f = f * i;  
    }  
    printf("The factorial is %d\n", f);  
    f = 1;  
}  
  
.*  
{  
    printf("Wrong Number\n");  
}  
%%  
int main()  
{  
    printf("Enter the number:\n");  
    yylex();  
    return 0;  
}  
int yywrap()  
{  
    return 1;  
}
```

▪ **Output:**

```
Enter the number:  
5  
The factorial is 120
```